Knowledge-Guided Vision-Language Models Prompt Learning for Open-Vocabulary Remote Sensing Image Scene Classification

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***Abstract*—Remote sensing open-vocabulary scene classification aims to recognize unseen scene categories beyond the predefined training set by leveraging external knowledge and vision-language models. Prompt learning has emerged as an effective paradigm to adapt pretrained vision-language models to open-vocabulary remote sensing image processing tasks by dynamically encoding task-specific knowledge into textual prompts. However, most existing approaches mainly rely on category names derived from training data, which lack fine-grained semantic descriptions and attribute-level associations that are crucial for complex remote sensing scenes. This limitation often degrades generalization when encountering novel classes and diverse geospatial domains. To address this problem, we propose KnowProKD, a knowledge-guided framework that incorporates diverse external knowledge and constraint-based regularization to enhance base-to-novel generalization. Specifically, KnowProKD consists of three components: 1) leveraging large language models (LLMs) to generate diverse attribute and description knowledge for constructing knowledge-aware prompts, improving cross-modal semantic alignment; 2) introducing knowledge-based constraints to regularize training and enhance robustness to novel category recognition; and 3) designing a dual-branch logits calibration mechanism that exploits attribute and description prompts to balance performance across base and novel classes. Extensive experiments on eight challenging remote sensing scene classification benchmarks demonstrate that KnowProKD achieves favorable performance compared with recent prompt-learning baselines and generalizes well to previously unseen scenarios. Furthermore, we apply KnowProKD to cross-regional urban land-use mapping and statistical analysis across representative cities from six continents, validating its practical effectiveness in real-world remote sensing applications.**

***Index Terms*—Generalization, knowledge-aware, prompt learning, remote sensing, scene classification.**

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I. Introduction

REMOTE sensing image scene classification involves the categorization of various land cover types and objects present in satellite or aerial imagery [1], [2], which is essential for numerous applications, including environmental monitoring [3] and urban planning [4], [5]. Traditional remote sensing image scene classification approaches predominantly depend on handcrafted features and conventional machine learning algorithms, which often exhibit limited generalization capabilities when confronted with diverse and complex scene variations [6], [7]. Open-vocabulary remote sensing image scene classification extends the capabilities of traditional models by enabling them to recognize categories beyond a fixed set of predefined categories [8]. This approach leverages large-scale vision-language models that are pretrained on extensive datasets containing a wide variety of image–text pairs. By utilizing remote sensing scene textual descriptions as prompts, these models possess the potential to generalize to novel unseen scene categories and scenarios, making them highly adaptable and robust for real-world applications.

The development of vision-language models, such as CLIP [9], ALIGN [10], FLIP [11], Qwen2-VL [12], and LLaVA [13] has significantly promoted open-vocabulary remote sensing image processing by effectively leveraging textual descriptions for zero/few-shot learning tasks [14], [15], [16], [17]. The ability to utilize these textual descriptions represents a critical advancement and research hotspot in this field. These models achieve their performance through an innovative pretraining approach that jointly optimizes image and text encoders using contrastive learning on massive collections of web-harvested image–text pairs. This sophisticated training paradigm facilitates the development of a nuanced comprehension of open-vocabulary visual concepts through robust cross-modal alignment, enabling effective semantic bridging between visual and textual domains. The models operate by employing carefully crafted textual prompts (e.g., "`a photograph of a <class>`") as queries to the text encoder, whose output embeddings are then compared with image encoder embeddings through similarity matching to determine class predictions. This versatile architecture has enabled these pretrained vision-language models to excel across a broad spectrum of tasks, including open-vocabulary

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image recognition [18], semantic segmentation [19], [20], [21], and object detection [22], marking a significant advancement in visual understanding systems. However, fully leveraging the potential of these models across diverse downstream remote sensing image processing tasks remains challenging, primarily because fixed pretrained representations often fail to adapt to the specific characteristics and domain shifts present in remote sensing data. Prompt learning has consequently emerged as a promising paradigm for enabling more flexible and task-adaptive model tuning.

Prompt learning approaches such as CoOp [23] and CoCoOp [24] introduce continuously learnable embedding vectors that, when combined with class-specific textual information, serve as prompts to adapt CLIP-based models for downstream tasks while preserving the pretrained model weights. VPT [25] learns prompts by designing visual prompts for CLIP's vision branch. ProGrad [26] updates prompt gradients to match the general knowledge from the original prompt. Despite the effectiveness of these auxiliary prompts, most previous works either rely solely on category names or retrieve information from existing knowledge bases like Wikipedia [27], [28], leading to two major drawbacks as follows.

1) Relying solely on remote sensing image scene category names provides limited class-related semantic knowledge, as category names are often too brief or abstract to fully capture the nuanced characteristics and contextual meanings of each class. Without additional descriptive information or external knowledge, the model may struggle to distinguish between semantically similar remote sensing image scene categories or understand the specific features that define them.

2) Knowledge retrieved from fixed databases often lacks the necessary diversity and domain specificity required for comprehensive understanding. Fixed databases are often constrained in scope and may not accurately represent real-world knowledge across diverse domains. Consequently, the information retrieved can be overly general or outdated, failing to capture the nuanced, context-specific insights that are crucial for a wide range of tasks.

The above limited knowledge scope provides insufficient guidance for learning, increasing the risk of overfitting to specific tasks. As a result, models often struggle to generalize to novel tasks or data distributions beyond those seen during training. Enhancing the informativeness and generalizability of learnable prompts thus remains a key challenge in developing more robust and adaptable vision-language prompt learning frameworks for open-vocabulary remote sensing image scene classification. To address these limitations, we propose a novel remote sensing image scene classification approach that leverages both attribute-level characteristics and conceptual descriptions by systematically incorporating external categorical knowledge into the prompt learning framework.

Through comprehensive experimental evaluations conducted across eight diverse remote sensing scene datasets, the proposed approach demonstrates consistent performance improvements over existing prompt learning methods. As depicted in Fig. 1, KnowProKD introduces several key components that distinguish it from previous approaches as follows.

1) *Diverse Remote Sensing Image Scene Knowledge Generation and Awareness for Prompt Learner:* Large language models (LLMs) are introduced to systematically generate diverse attribute-level information and comprehensive descriptions for each remote sensing image scene category. For this knowledge (categories, attributes, and descriptions), we design dedicated learnable embedding representations as prompt learners, which are integrated into the vision-language branch through knowledge-aware prompt learning, enabling the model to better perceive and be aware of remote sensing image scene categories by utilizing both common external descriptions and fine-grained attribute features on remote sensing image scenes.
2) *Knowledge Constraints for Novel Remote Sensing Image Scene Classification:* While conventional prompt learning with remote sensing image scene category information enables vision-language models to adapt to specific tasks, it often comes at the cost of losing the valuable robustness capabilities present in the original pretrained models, particularly affecting performance on novel unseen categories. To address this, we introduce a constraint mechanism for remote sensing image scene knowledge prompt learning that leverages external attribute and descriptive knowledge features to regularize the learnable prompts associated with categories. The inherent diversity of this knowledge improves the model's robustness across novel unseen remote sensing image scene categories.
3) *Knowledge Prompt Logits Calibration for Diverse Remote Sensing Scene Classification:* Existing class-name prompt learning methods mainly compute classification logits from the similarity between image embeddings and class textual embeddings, which may be insufficient for remote sensing scenes with high intraclass diversity and strong interclass ambiguity. To improve the reliability of open-vocabulary predictions, we introduce two auxiliary knowledge prompt branches based on category attributes and general descriptions, which provide complementary evidence to the class-prompt logits. We then perform a lightweight logits calibration by fusing the class, attribute, and description logits with fixed weights, yielding more stable and discriminative decision scores. This design enhances prediction robustness on novel categories and improves the overall generalization performance in diverse remote sensing scene classification.

Extensive experiments conducted on eight remote sensing scene datasets demonstrate the effectiveness of the proposed method. Our approach achieves favorable results compared with recent prompt-learning baselines across multiple metrics and evaluation settings. Moreover, ablation studies analyze the contribution of each component in the proposed framework. The results show the individual effects of knowledge-aware prompt learning, knowledge constraints, and logit calibration. Furthermore, we demonstrate the practical utility of our approach through a cross-region urban land-use mapping and

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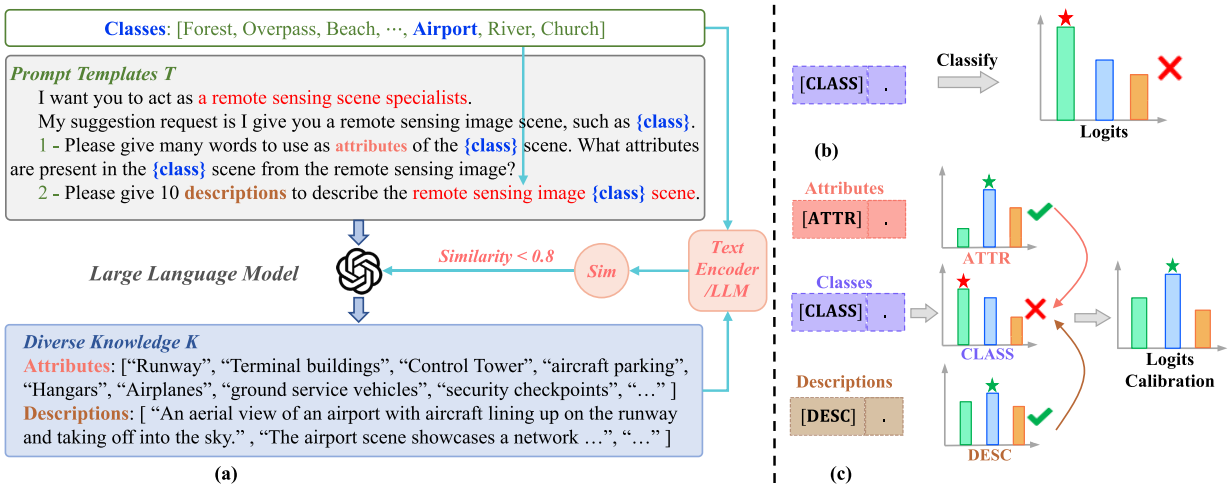


Fig. 1. Comparison of KnowProKD with standard prompt learning methods. (a) Generating scene knowledge with an LLM. (b) Standard prompt learning without external scene knowledge. (c) Using scene knowledge to calibrate results of prompt learning.

statistics analysis case study, which validates the model's effectiveness in real-world remote sensing applications.

The rest of this article is organized as follows. Section II reviews related work on open-vocabulary scene classification, vision-language models, and prompt learning. Section III details the proposed KnowProKD framework, covering knowledge generation, knowledge-aware prompt learning, and knowledge-based logits calibration. Section IV presents experimental results, analysis, and applications on multiple remote sensing datasets. Section V concludes this article and outlines future directions.

II. Related Work

A. Open-Vocabulary Remote Sensing Image Scene Classification

Remote sensing scene image classification has traditionally been dominated by supervised learning paradigms, which rely on large-scale labeled datasets with predefined closed-set categories [29]. While these models achieve high accuracy on seen classes, their performance degrades significantly when encountering novel scene types, making them less adaptable to dynamic and diverse remote sensing environments. To address this limitation, open-vocabulary classification has emerged as a promising direction, aiming to enable models to recognize unseen categories by leveraging auxiliary knowledge sources, such as vision-language models, self-supervised pretraining, and prompt learning. Recent advancements have explored the adaptation of contrastive language-image pretraining [9] and similar VLMs to remote sensing data, bridging the gap between natural image understanding and aerial scene recognition. Techniques such as feature alignment, prompt engineering, and cross-modal knowledge distillation have been employed to enhance the transferability of visual and textual embeddings from general-purpose models to the remote sensing domain. In addition, hybrid approaches that integrate graph-based semantic reasoning, meta-learning, and domain adaptation have been proposed to mitigate domain shift issues and improve generalization to unseen categories.

B. Vision-Language Models

Recently, there has been a significant trend toward combining images and text for self-supervised pretraining of vision-language large models that encode multimodal representations of visual and textual modalities. Vision-language models such as ALIGN [10], CLIP [9], and CoCa [30] have been widely applied to various tasks, including zero-shot and open-vocabulary visual recognition, demonstrating strong performance across diverse applications. These vision-language models are mainly dual-stream encoder architectures consisting of a text encoder and an image encoder, which are pretrained using a large number of image–text pairs on the internet to combine image languages in a self-supervised manner through contrastive learning. Although these pretrained vision-language models learn generalized modal representations, applying them to more downstream tasks remains a challenging problem. Many works introduce task-specific modifications to vision-language models to make them adaptable to downstream tasks. In this work, we propose a diverse knowledge-injected pretrained model fine-tuning technique to effectively make vision-language models adaptable to few-shot and zero-shot visual recognition tasks.

C. Prompt Learning

Prompt learning originally came from the field of NLP. In order to adapt the pretrained language model to downstream tasks by adding additional language instructions to the text, it has achieved good results in NLP tasks. Recently, the field of computer vision has also intensively studied prompt learning for solving downstream tasks. Manually designed prompt templates in the original CLIP have been used in vision-language tasks. In order to design more comprehensive, robust prompts, CoOp [23] fine-tunes CLIP for image recognition by designing continuous learnable prompt embeddings in the text encoder of its language branch. Simple learnable context prompts have weak generalization; CoCoOp [24] further generates learnable conditional prompts for images through a lightweight meta-net, improving the generalization

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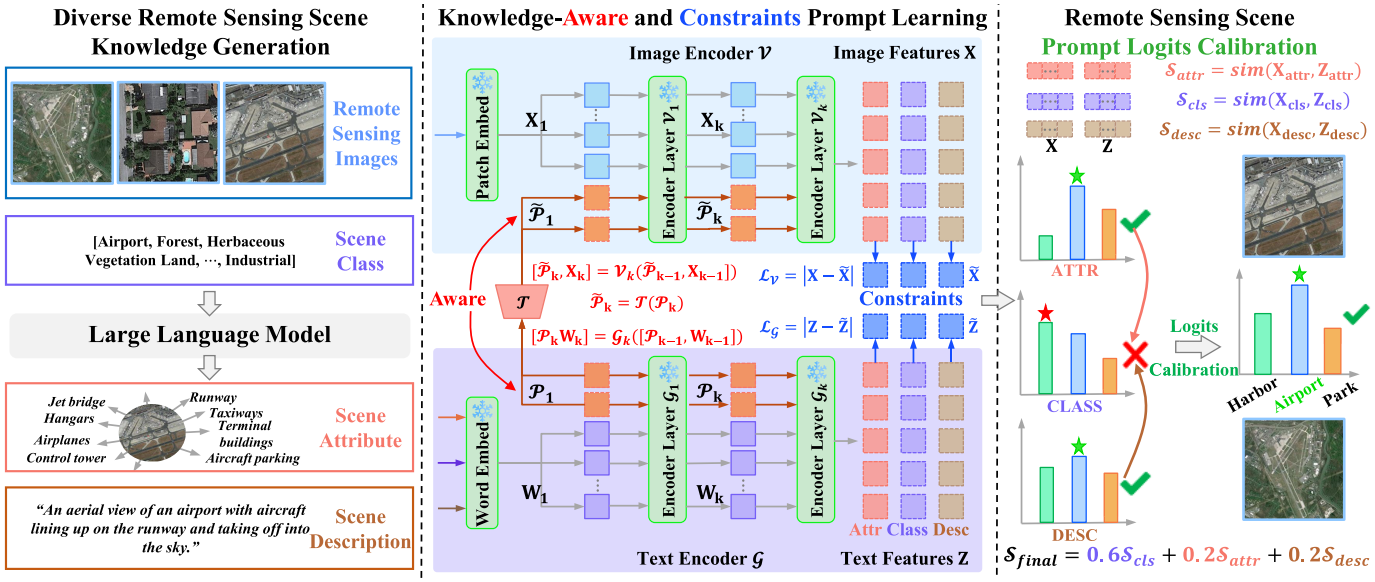


Fig. 2. Overview of our proposed KnowProKD framework for prompt learning in vision-language models. It consists of three parts: knowledge generation, fine-tuning the model with knowledge-aware prompts and knowledge constraints, and calibrating the logits of knowledge prompts. KnowProKD fine-tunes the vision-language model based on the generated external diverse knowledge construction attributes, general descriptions, category-aware prompts, and knowledge constraints on prompt features. Finally, using the attribute and description prompt features, the probabilities calculated from the category prompt are corrected to increase the generalization and robustness of the model.

performance of the task. VPT [25] performs prompt learning by designing visual prompts for the vision branch of CLIP. ProGrad [26] updates the gradient of the prompt to align it with the generalized knowledge generated by the original prompt. MaPLe [31] proposes a vision-language multimodal prompt learning method that performs joint learning by designing prompts on the multilayer structure of CLIP's visual and language encoders to improve the alignment between visual and language representations. The proposed method performs joint learning by designing multiple types of external knowledge prompts and injecting them into the visual and language encoders.

To further enrich prompt diversity and semantic expressiveness, a growing body of work has explored the integration of external knowledge sources such as Wikipedia, ConceptNet [32], and WordNet [33] to support downstream task performance. In visual question answering (VQA), the incorporation of structured knowledge graphs and encyclopedic facts has proven effective in improving reasoning and factual grounding. Recent efforts have extended this idea to prompt learning: for example, one line of work leverages Wikipedia to construct two complementary types of knowledge prompts, showing improved generalization to novel classes, though with limited knowledge diversity. Another approach, K-LITE [34], augments textual entities with external structured knowledge to support efficient and scalable zero-shot and few-shot transfer. In contrast to approaches that depend on static external knowledge bases, we propose to generate diverse knowledge prompts directly via LLMs. These LLM-generated prompts are adaptable to diverse scenarios and overcome the limitations of retrieval-based pipelines, offering a more flexible and scalable approach to improving vision-language prompt learning.

Recent prompt learning studies also improve adaptation from different perspectives. DPLQ [35] focuses on domain prompt learning and representation enhancement with quaternion networks for domain transfer, while PromptKD [36] transfers knowledge from a teacher model to a student model through unsupervised prompt distillation. These methods do not explicitly generate category-level attribute and description knowledge for each remote sensing scene class. In contrast, KnowProKD constructs LLM-generated scene knowledge and uses it as explicit semantic guidance in prompt construction, training regularization, and prediction calibration.

III. Approach

This study presents a diverse knowledge-injection strategy for vision-language modeling within the prompt learning paradigm. Built upon the multimodal CLIP framework, the proposed approach improves generalization to downstream tasks through prompt context optimization. Fig. 2 illustrates the overall architecture of KnowProKD. Unlike prior methods that rely solely on category information for prompt fine-tuning, KnowProKD incorporates diverse external knowledge, including attributes and general descriptions, to construct knowledge-aware prompts with multiple learnable contexts. This design enables multidimensional adaptation of the vision-language model and enhances its generalization and robustness. Compared with recent methods such as DPLQ and PromptKD, the key distinction of KnowProKD lies in how external knowledge is obtained and used. Rather than relying only on domain-specific prompt parameterization or teacher–student prompt distillation, KnowProKD first generates category-level remote sensing scene knowledge with an LLM and then incorporates the generated attributes and

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descriptions into three parts of the framework: knowledge-aware prompt construction, knowledge-based regularization, and logits calibration during inference. Thus, the generated scene knowledge serves not only as auxiliary textual information, but also as explicit semantic guidance for representation learning and prediction calibration.

A. Knowledge Generation Pipeline

Information and knowledge directly derived from the image itself are inherently limited in scope and primarily reflect image-specific semantics. Such information is commonly used in prompt learning to construct learnable contextual embeddings by associating image features with category names. However, this strategy restricts the prompts to rely on low-level visual cues and lacks access to richer, high-level semantic concepts that are often crucial for robust and transferable visual understanding. To address this limitation, we propose to incorporate external category-related knowledge that is not confined to the image itself but rather reflects the broader semantic context of the category. Specifically, we leverage one of the most advanced LLMs, GPT, to automatically extract diverse, generalized knowledge associated with each category. This knowledge serves to enrich the textual context used in prompt tuning, thereby bridging the gap between visual and semantic understanding.

Utilizing LLMs in this context offers several key advantages as follows.

1) For any given dataset, structured and interpretable knowledge, such as scene attributes and general category descriptions, can be generated automatically by designing domain-agnostic prompt templates, eliminating the need for manual annotation.

2) LLMs are capable of producing semantically rich and diverse textual representations, offering comprehensive and fine-grained descriptions of category characteristics beyond what is observable in a single image.

3) Since both CLIP and LLMs are trained on large-scale web corpora, the textual knowledge generated by LLMs is naturally aligned with CLIP's learned semantic space, ensuring compatibility and enhancing the model's capacity to represent and distinguish categories more effectively.

This synergy between LLM-generated semantic knowledge and CLIP's vision-language alignment contributes to more robust prompt representations, ultimately leading to improved generalization across diverse image domains.

We construct domain-specific prompt templates T based on remote sensing scene datasets from different domains and employ an LLM to generate two complementary types of category-level knowledge for each class C, including attribute knowledge $\mathcal{A}$ and general description knowledge $\mathcal{D}$. To improve the reproducibility of the LLM-based knowledge generation process, we further clarify the refinement strategy used in our pipeline. For each category, the LLM is instructed to generate multiple attribute words as well as ten diverse descriptive sentences following the prompt template T, as illustrated in Fig. 1(a). To remove low-relevance generated candidates, we perform an iterative similarity-based relevance filtering procedure. Specifically, the similarity is computed between each newly generated candidate and its original category name, rather than between the candidate and the existing knowledge pool. Both texts are encoded by the `bge-m3` text embedding model [37], and their cosine similarity is computed. Candidates with Sim < 0.8 are regarded as insufficiently relevant to the target category and are discarded and resampled. This refinement process is repeated until the required number of attribute and description knowledge items is obtained for each category. Finally, the filtered knowledge sets are organized as $\mathcal{A} = \{[a_1], [a_2], \ldots, [a_N]\}$ and $\mathcal{D} = \{[d_1], [d_2], \ldots, [d_N]\}$, and the overall knowledge generation process is formulated as

$$[\mathcal{A}, \mathcal{D}] = LLM(C, T). \tag{1}$$

The generated external knowledge $\mathcal{A}$ and $\mathcal{D}$ provides richer semantic cues beyond category names, thereby supporting knowledge-aware prompt construction for effective prompt learning in vision-language models.

B. Diverse Knowledge-Aware and Constraints

Despite the significant zero-shot capabilities of CLIP, we observe a fundamental limitation in existing prompt learning approaches applied to downstream tasks: they typically rely solely on class labels or category names to construct prompts. While category-level information provides a basic semantic anchor, it is often too coarse and lacks sufficient contextual richness to facilitate effective knowledge transfer. As a result, these prompt representations exhibit limited expressiveness, leading to suboptimal generalization—particularly in scenarios involving novel domains, unseen categories, or significant domain shifts. This limitation becomes especially pronounced in multidomain or open-set visual recognition settings, where semantic ambiguity and category variability are common. In such cases, prompts constructed purely from category names fail to capture the diverse contextual cues that are critical for robust recognition and adaptation.

To address this challenge and better leverage CLIP's powerful vision-language alignment for downstream tasks, we propose a knowledge-enhanced prompt learning framework that integrates rich external knowledge into the prompt construction process. Specifically, we incorporate fine-grained scene attributes, such as objects, textures, or spatial layouts, along with generalized category descriptions obtained from LLMs. These diverse sources of external knowledge are used to generate knowledge-aware prompts that go beyond simple category names and encode deeper semantic information.

The integration of such diverse knowledge serves a dual purpose as follows.

1) It enhances the model's perception and discrimination ability by providing multilevel semantic context, thereby improving recognition of both base and novel categories.

2) It enables the formulation of knowledge-guided constraints during fine-tuning, where the scene attributes and descriptions act as regularizers to align the learning trajectory with more generalized representations. Through this strategy, our framework significantly boosts the robustness and generalization capacity of the vision-language model across a wide range of downstream visual domains and classification tasks.

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1) Diverse Knowledge-Aware Prompt Learning: For open-vocabulary image classification, CLIP encodes an input image into an image embedding $\mathbf{x}$. Meanwhile, it constructs a set of class-specific textual prompts, e.g., `a photograph of a<class>`, and feeds them into the text encoder to obtain the corresponding text embeddings. Assuming that the total number of categories is C, we denote the text embedding of the ith class prompt as $\mathbf{z}_i$. CLIP then computes the classification probability for the ith class by applying softmax over cosine similarities between the image embedding and all class text embeddings

$$p(y=i|\mathbf{x}) = \frac{\exp\left(\text{sim}\left(\mathbf{x}, z_i\right)/\tau\right)}{\sum_{j=1}^{C}\exp\left(\text{sim}\left(\mathbf{x}, z_j\right)/\tau\right)} \tag{2}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity and τ is the temperature hyperparameter. Note that $\mathbf{z}_i$ represents the standard CLIP text embedding derived from vanilla class prompts. In the following, we use $\mathbf{z}_{kg}$ to denote the knowledge-enhanced text embeddings obtained from our diverse knowledge-aware prompts, including category, attribute, and description prompts.

In order to fully utilize the external multiple knowledge sources, we design a series of knowledge-aware prompts for fine-tuning CLIP. By utilizing attribute and general description knowledge to construct learnable contextual attribute knowledge-aware tokens P_{attr}, P_{desc}, while preserving the original category information of the prompts P_{cls}. So that there are $P_{kg} = \{P_{attr}, P_{cls}, P_{desc}\}$. The input embeddings is $[P_{kg}, W_{kg}^0]$, where $W_{kg}^0 = [w^1, w^2, \ldots, w^N]$ corresponds to fixed language tokens, with the subscripts "kg" corresponding to the input the category C, attribute $\mathcal{A}$, and general description $\mathcal{D}$ knowledge. New learnable tokens are further introduced in each transformer block of the language encoder ($\mathcal{G}_i$) and the visual encoder ($\mathcal{V}_i$) up to a specific depth m. The new learnable tokens are then used in the transformer block

$$\left[_,\ W^i\right] = \mathcal{G}_i\left(\left[P_{kg}^{i-1}, W^{i-1}\right]\right) \qquad i = 1, 2, \ldots, m. \tag{3}$$

In this context, the concatenation operation is denoted by $[\cdot,\cdot]$. After the mth transformer layer, the subsequent layers are dedicated to processing prompts from the preceding layer, ultimately culminating in the computation of the final textual embedding z_{kg}

$$\left[P_{kg}^j,\ W_{kg}^j\right] = \mathcal{G}_j\left(\left[P_{kg}^{j-1}, W_{kg}^{j-1}\right]\right) \qquad j = m+1, \ldots, h \tag{4}$$

$$\mathrm{z}_{kg} = \texttt{TextProj}\left(w_h^N\right). \tag{5}$$

As illustrated in Fig. 3, we embed a set of knowledge-aware learnable visual prompts $\tilde{P}_{kg}^i = \{\tilde{P}_{attr}^i, \tilde{P}_{cls}^i, \tilde{P}_{desc}^i\}$ into CLIP's vision branch to harness the synergistic potential of vision-language alignment. These prompts are generated by projecting the language-based knowledge prompts P_{kg}^i via a cross-modal projection module, denoted by the vision-language coupling function $\mathcal{T}$. Prompt learning is an effective way to adapt the pretrained vision-language model features to the downstream task, with $\tilde{P}_{kg}^i = \mathcal{T}(P_{kg}^i)$

$$\left[v_{kg}^i, E_{kg}^i,\ _\ \right] = \mathcal{V}_i\left(\left[v_{kg}^{i-1}, E_{kg}^{i-1}, \tilde{P}_{kg}^{i-1}\right]\right) \tag{6}$$

$$\left[v_{kg}^j, E_{kg}^j, \tilde{P}_{kg}^j\right] = \mathcal{V}_j\left(\left[v_{kg}^{j-1}, E_{kg}^{j-1}, \tilde{P}_{kg}^{j-1}\right]\right) \tag{7}$$

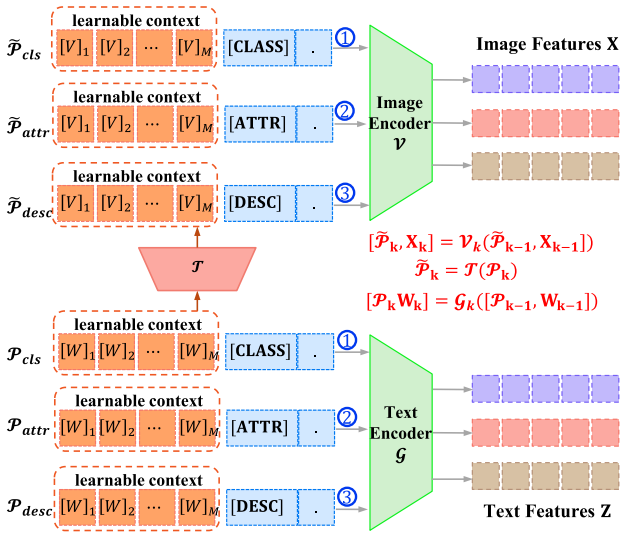


Fig. 3. Illustration of the proposed diverse knowledge-aware prompt learning method.

$$i = 1, \ldots, m; \qquad j = m+1, \ldots, h \tag{8}$$

$$\mathrm{x}_{kg} = \texttt{ImageProj}\left(v_h\right) \tag{9}$$

where v_{kg}^i denotes the visual token representations at the ith transformer block of the vision encoder under the knowledge-aware prompt setting, and E_{kg}^i denotes the corresponding image patch embedding tokens propagated through the same block. The learnable visual prompts $\tilde{P}_{kg}^i$ are inserted into the vision branch as additional prompt tokens, including category, attribute, and description prompts. Here, i and j index the transformer blocks, m indicates the depth at which the prompt tokens are injected, and h denotes the total number of transformer blocks in the vision encoder.

By using category, attribute, and general description knowledge for prompt learning, the three types of knowledge-aware prompts are separately optimized using cross-entropy loss $\mathcal{L}_{\text{CE-kg}}$, which can be uniformly represented as

$$\mathcal{L}_{\text{CE-kg}} = \arg\min_{P_{kg}} \mathbb{E}_{(X,y)\sim\mathcal{D}}\,\mathcal{L}\left(\text{sim}\left(\mathrm{z}_{kg}, \mathrm{x}_{\text{kg}}\right), y\right). \tag{10}$$

Here, $\text{sim}(\cdot,\cdot)$ denotes cosine similarity. Before computing the similarity, both image and text features are ℓ_2-normalized so that the similarity score is bounded within $[-1, 1]$. A larger similarity value indicates stronger cross-modal alignment between the image representation and the corresponding textual prompt embedding.

2) Diverse Knowledge Constraints: The cross-entropy loss is employed to optimize the model using prompts trained on the base classes, thereby encouraging the prompt representations to capture discriminative knowledge specific to those categories. However, as training progresses over multiple iterations, the learned prompts are prone to overfitting, focusing excessively on the task-specific distributions seen during training. This overfitting diminishes the capacity of the prompts to effectively leverage the generalized, domain-agnostic knowledge encoded in the frozen CLIP model.

We introduce a novel constraint mechanism that regularizes the optimization of category prompts using three types of frozen features: attribute-level embeddings, general descriptive

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knowledge, and textual representations of class semantics. By incorporating these fixed, high-level knowledge sources as auxiliary guidance, we enforce the learned prompts to remain aligned with the broader conceptual space encoded in CLIP, rather than deviating toward narrow task-specific solutions. These external attribute features and descriptive text embeddings serve as generalized representations of each category, independent of the specific base classes or training dataset. By integrating them into the optimization process, we can continuously steer the learning trajectory of the category prompts toward representations that are semantically rich and transferable. This not only reduces overfitting on base classes but also improves the model's generalization to novel categories and unseen domains. Category prompts are essentially learnable token embeddings combined with class names, and their parameters are iteratively updated during training. Thus, it becomes crucial to anchor their optimization using frozen class text features as reference features. Similarly, we constrain the learning of visual class prompts by aligning the generated image features with the frozen image representations from CLIP. This dual-branch regularization framework enhances consistency between learned prompts and CLIP's pretrained knowledge. The overall objective is formalized as follows:

$$\mathcal{L}_{\text{KC-image}} = \sum_{i=1}^{d} \left| \mathrm{x}_{cls} - \tilde{\mathrm{x}} \right|, \qquad \mathcal{L}_{\text{KC-text}} = \sum_{i=1}^{d} \left| \mathrm{z}_{kg} - \tilde{\mathrm{z}}_{kg} \right| \quad (11)$$

where $\mathrm{x}_{cls}, \mathrm{z}_{kg} \in \mathbb{R}^d$ are the prompt-induced image and text embeddings, and $\tilde{\mathrm{x}}, \tilde{\mathrm{z}}_{kg} \in \mathbb{R}^d$ denote the corresponding frozen CLIP reference features. Here, d is the feature dimension, and the constraint terms are implemented as an ℓ_1 penalty to suppress representation drift.

The proposed diverse knowledge constraints are introduced as a regularization term to prevent the learnable prompts from overfitting to base classes during prompt tuning. Concretely, the constraints penalize the deviation between the prompt-induced representations and the corresponding frozen CLIP text/image features, thereby anchoring optimization to the pretrained vision-language semantic space. This design encourages semantic consistency and reduces representation drift, which improves robustness on novel categories.

Regarding convergence, the overall objective is the sum of the standard cross-entropy loss and differentiable constraint terms, and it can be optimized jointly with standard SGD under the same training setting as conventional prompt learning. In practice, updating only lightweight prompt parameters while keeping the CLIP backbone frozen leads to stable empirical convergence across all datasets.

C. Vision Language Prompt Coupling

Inspired by MaPLe [31], which couples learnable prompts across the vision and language branches for improved multimodal adaptation, we introduce a vision-language prompt coupling mechanism tailored to knowledge-enhanced prompt learning. The key motivation is to transfer knowledge-aware prompt information learned in the language branch into the vision branch, so that both modalities can be jointly optimized under consistent semantic guidance, which is particularly beneficial for open-vocabulary recognition under domain shift.

Specifically, our framework contains three types of knowledge-aware prompts, including category prompts, attribute prompts, and description prompts. Let $P_{kg}^i = \{P_{cls}^i, P_{attr}^i, P_{desc}^i\}$ denote the learnable knowledge prompts at the ith transformer block of the text encoder. We employ a lightweight cross-modal projection module to map these language-side prompts into the vision branch and obtain the corresponding visual prompts $\tilde{P}_{kg}^i = \{\tilde{P}_{cls}^i, \tilde{P}_{attr}^i, \tilde{P}_{desc}^i\}$. This coupling enables knowledge cues at different semantic levels to be injected into the visual feature extraction process, encouraging better cross-modal alignment. In practice, the coupling function $\mathcal{T}$ is implemented as a learnable projection that operates on the prompt embeddings, and the projected visual prompts are inserted into the vision transformer blocks for prompt tuning. By coupling category, attribute, and description prompts in a unified manner, the proposed design provides multilevel semantic guidance from language to vision, thereby improving robustness and generalization to novel remote sensing scene categories.

D. Diverse Knowledge Logits Calibration

In general prompt learning frameworks, classification is typically achieved by fine-tuning the model using learnable class prompts, followed by the extraction of image and text features. These features are then compared through similarity metrics to compute the classification logits for each category. However, this process heavily relies on the relevance between the visual content and the class-level prompt representation. When the semantic alignment between the input image and the textual class prompt is weak, particularly in cases involving visually complex scenes, domain shifts, or unfamiliar categories, the model is prone to generating unreliable logits, leading to misclassifications and reduced robustness.

To alleviate this issue and improve the model's fault tolerance in low-relevance or ambiguous scenarios, we propose a logits calibration mechanism that integrates knowledge-aware prompt guidance into the classification process. Specifically, while retaining the original learnable class prompts, we augment the prompt space with two additional types of knowledge-enhanced prompts: 1) attribute-level prompts, which capture fine-grained visual properties (e.g., color, shape, and texture) and 2) general description prompts, which encode high-level semantic descriptions of the category. Each type of prompt is independently trained to encode meaningful external knowledge. During inference, we extract both image and text features using the class, attribute, and description prompts. For each of these three branches, similarity scores between visual and textual embeddings are computed to produce individual sets of classification logits.

To synthesize these outputs and reduce over-reliance on any single modality or representation, we propose a logit fusion strategy. In this strategy, the logits derived from the class prompts are calibrated using those from the attribute and general description prompts. These knowledge-guided logits serve as dominant signals, acting as soft constraints to adjust

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and refine the initial predictions made by class prompts. The final classification output is obtained through a weighted fusion of all three branches, resulting in the calibrated logits $\mathcal{P}_{final}$, formally defined as

$$\mathcal{P}_{kg}\left(y=i|\mathbf{x}_{kg}\right)=\frac{\exp\left(\text{sim}\left(\mathbf{x}_{kg},z_{kg}^{i}\right)/\tau\right)}{\Sigma_{j=1}^{C}\exp\left(\text{sim}\left(\mathbf{x}_{kg},z_{kg}^{j}\right)/\tau\right)} \tag{12}$$

$$\mathcal{P}_{final}=\mathcal{P}_{cls}+\lambda_1\mathcal{P}_{attr}+\lambda_2\mathcal{P}_{desc} \tag{13}$$

where $\mathcal{P}_{cls}$ denotes the primary classification logits, and λ_1 and λ_2 are fixed balancing coefficients for the attribute and description logits, respectively. In all main experiments, we set $\lambda_1=\lambda_2=0.2$.

Through the integration of diverse knowledge sources and prompt learning, our logits calibration module significantly enhances the semantic alignment between visual and textual representations. By leveraging external knowledge to guide the calibration process, the module effectively refines the classification logits produced by class prompts, leading to more accurate and robust scene recognition performance across both base and novel categories.

IV. Experiments

To rigorously evaluate the proposed method, we conducted three experiments: generalization from base-to-novel classes, cross-dataset evaluation, and few-shot experiments. The proposed method KnowProKD achieves competitive performance across eight remote sensing scene datasets, highlighting its efficacy in recognizing novel categories, facilitating cross-domain generalization, and addressing limited-data scenarios.

A. Experiment Setup

1) Datasets: The base-to-new generalization and few-shot experiments are conducted on eight remote sensing scene datasets, i.e., MLRSNet [38], NWPU-RESISC45 [1], PatternNet [39], UCM [40], AID [41], WHURS19 [42], RSSCN7 [43], and RSICD [44]. The cross-dataset evaluation is performed by prompt tuning on MLRSNet and evaluating on the other seven remote sensing scene datasets.

2) Knowledge Generation Details: We provide the detailed settings of the LLM-based knowledge generation process for reproducibility. OpenAI GPT-4.5 [45] is used to generate two types of category-level knowledge for each scene class, including attribute knowledge $\mathcal{A}$ and general description knowledge $\mathcal{D}$. For each class, $N=10$ attribute candidates and $N=10$ descriptive sentences are generated. The default sampling configuration is adopted with temperature set to 0.7 and top-p set to 0.9, which encourages knowledge diversity while preserving semantic relevance. A similarity-based refinement strategy is applied to filter low-relevance candidates. Specifically, the similarity is computed between each generated candidate and the original category name, rather than between the candidate and the existing knowledge pool. Both texts are encoded using the `bge-m3` text embedding model [37], and cosine similarity is computed. Candidates with Sim <0.8 are discarded and resampled. The refinement is repeated until the target number of knowledge items is obtained for each category, and the filtered knowledge is used to construct knowledge-aware prompts for subsequent prompt tuning. The LLM is used only once to build an offline category-level knowledge bank before prompt tuning, and the filtered knowledge is fixed during all training and testing stages. Therefore, LLM sampling does not introduce additional randomness into model training or inference. Although different random seeds may lead to lexical variations, such as synonyms or slightly different sentence structures, the relevance filtering step encourages the retained knowledge to describe the same core scene semantics. Moreover, using multiple attributes and descriptions for each category reduces the influence of any single generated expression, making the prompt knowledge reasonably stable for the downstream experiments.

B. Generalization From Base-to-Novel Classes

We evaluate KnowProKD against recent approaches, including zero-shot CLIP [9], CoOp [23], CoCoOp [24], ProGrad [26], MaPLe [31], PromptSRC [46], CoPrompt [47], DPLQ [35], and PromptKD [36]. Our evaluation covers eight diverse remote sensing scene datasets, providing a broad comparison under the base-to-novel setting. The experimental results in Table I demonstrate clear performance improvements of the proposed method over existing approaches on remote sensing scene classification. A key observation is that methods such as CoOp, CoCoOp, and ProGrad often show reduced novel-class performance compared with the zero-shot CLIP baseline after prompt learning. This suggests that category-name prompt tuning may overfit base classes and weaken generalization to novel categories. Recent prompt learning methods alleviate this issue to some extent, while KnowProKD further improves novel-class recognition by incorporating diverse knowledge evidence into prompt learning and inference-time prediction refinement.

Our proposed KnowProKD improves open-vocabulary remote sensing scene classification by incorporating diverse knowledge evidence into inference-time prediction refinement. Table I reports the base-to-novel generalization results on eight remote sensing scene datasets. On the average of all datasets, KnowProKD achieves 96.39% base accuracy, 82.91% novel accuracy, and 89.14% harmonic mean (HM), outperforming the previous prompt-learning baseline MaPLe by + 3.27% on base accuracy, + 11.20% on novel accuracy, and + 8.72% on HM. Compared with the recent knowledge-enhanced method DPLQ, KnowProKD further improves the average HM from 83.50% to 89.14%, yielding a gain of + 5.64%. Across individual datasets, KnowProKD consistently achieves strong performance on both base and novel categories. For example, on NWPU-RESISC45, KnowProKD improves HM from 80.71% with MaPLe to 89.83%, and raises novel accuracy from 72.70% to 83.93%. On PatternNet, KnowProKD obtains 91.11% HM, surpassing MaPLe by + 19.08%. Fig. 4 provides a dataset-wise visualization of the harmonic-mean comparison and shows that KnowProKD achieves the highest HM on all eight datasets. These results indicate that incorporating diverse knowledge cues provides complementary evidence for correcting class-level predictions, which improves robustness

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TABLE I

COMPARISON WITH RECENT METHODS IN BASE-TO-NOVEL GENERALIZATION ON EIGHT REMOTE SENSING SCENE DATASETS. KNOWPROKD ACHIEVES FAVORABLE RESULTS ACROSS THE EVALUATED DATASETS

Dataset		CLIP	CoOp	CoCoOp	ProGrad	MaPLe	PromptSRC	CoPrompt	DPLQ	PromptKD	KnowProKD
Average on 8 datasets	Base	71.19	87.61	91.82	91.24	93.12	92.60	94.70	94.08	95.15	**96.39**
	Novel	71.33	70.84	68.98	70.98	71.71	72.97	74.64	75.06	79.57	**82.91**
	HM	70.63	78.03	78.43	79.85	80.42	81.62	83.48	83.50	86.67	**89.14**
NWPU-RESISC45	Base	69.00	84.53	89.27	88.97	90.70	91.77	93.83	91.70	96.60	**96.63**
	Novel	63.00	66.97	69.37	70.01	72.70	73.20	72.53	75.03	80.20	**83.93**
	HM	65.87	74.73	78.07	78.36	80.71	81.44	81.82	82.53	87.64	**89.83**
AID	Base	73.50	87.63	92.63	91.35	92.73	92.60	95.27	94.50	**96.83**	96.30
	Novel	70.40	70.37	65.73	68.65	74.57	69.27	74.77	75.77	77.10	**78.60**
	HM	71.92	78.06	76.89	78.39	82.66	79.25	83.78	84.10	85.85	**86.55**
PatternNet	Base	70.60	87.30	93.70	90.12	95.30	93.10	95.07	95.80	95.53	**96.77**
	Novel	62.60	64.20	59.90	59.70	57.90	66.77	63.63	66.20	85.03	**86.07**
	HM	66.36	73.99	73.08	71.82	72.03	77.77	76.24	78.30	89.97	**91.11**
UCM	Base	80.60	93.60	95.23	96.87	97.70	97.43	**99.23**	97.90	93.17	98.93
	Novel	68.00	74.53	71.57	70.00	70.90	72.00	73.00	73.30	77.00	**78.50**
	HM	73.77	82.98	81.27	81.27	82.17	82.81	84.12	83.83	84.32	**87.54**
MLRSNet	Base	64.50	79.37	83.30	83.23	85.23	85.97	87.67	87.07	91.57	**92.27**
	Novel	60.30	58.90	59.50	58.73	59.60	59.07	60.97	59.00	67.03	**69.20**
	HM	62.33	67.62	69.42	68.87	70.15	70.03	71.92	70.34	77.40	**79.09**
RSSCN7	Base	66.70	84.80	90.97	90.83	91.67	90.73	92.53	91.20	93.10	**95.57**
	Novel	95.30	89.13	90.00	94.17	93.70	90.80	93.70	95.57	90.20	**95.83**
	HM	78.48	86.91	90.48	95.63	92.67	95.63	93.11	93.33	91.63	**95.70**
RSICD	Base	71.50	88.43	92.37	91.23	93.93	93.63	95.67	95.67	96.23	**97.37**
	Novel	60.20	60.20	58.80	63.03	56.27	61.57	66.50	64.83	72.50	**76.03**
	HM	65.37	71.63	71.86	74.55	70.38	74.29	78.46	77.29	82.70	**85.39**
WHURS19	Base	73.10	95.20	97.10	97.30	97.70	95.60	98.33	**98.80**	98.17	97.30
	Novel	90.80	82.40	77.00	83.57	88.03	91.07	92.03	90.80	87.53	**95.10**
	HM	80.99	88.34	85.89	89.91	92.61	93.28	95.08	94.63	92.55	**96.19**

under domain shift and enhances recognition performance on novel categories.

C. Cross-Dataset Evaluation

To evaluate the cross-dataset generalization capability of the proposed diverse knowledge-aware prompt learning method, we conduct transfer experiments across multiple remote sensing scene datasets. Specifically, the model is first trained on the NWPU dataset, which contains 45 scene classes, and is then directly transferred to four additional remote sensing scene datasets, including EuroSAT. This setting is challenging because the target datasets differ from the source dataset in image resolution, geographic coverage, scene composition, and category distribution. Therefore, it provides a direct test of whether the learned knowledge-aware prompts can generalize beyond the source-domain training classes and imaging conditions.

As shown in Table II, KnowProKD achieves favorable transfer performance across the target datasets. In particular, it improves the average accuracy to 78.72%, exceeding CoPrompt by 2.84%. The gains are not limited to the source dataset, but are also observed on several target datasets, indicating that the introduced attribute and description knowledge provides complementary semantic cues for cross-dataset transfer. These results suggest that incorporating knowledge-aware prompts and designed constraints can improve cross-dataset generalization for remote sensing scene understanding.

D. Domain Generalization Experiments

To further validate the generalization ability of our model, we follow the previous study of evaluation under the domain generalization setting. Table III summarizes the results of KnowProKD and previous methods on out-of-distribution datasets. We evaluate the direct transferability of ImageNet-trained models to various out-of-distribution datasets, and our method consistently outperforms existing methods. These results show that our proposed diverse knowledge-aware and constrained prompt learning can effectively improve the generalization and robustness of vision-language models.

E. Few-Shot Experiments

To rigorously evaluate the robustness and adaptability of the proposed KnowProKD, we conduct comprehensive few-shot experiments across multiple remote sensing scene datasets. Performance is systematically examined over training regimes ranging from 1-shot to 16-shot, encompassing varying degrees of data availability and providing a thorough assessment of the framework's generalization capability in data-limited scenarios.

As demonstrated in Fig. 5, the comparison across different shot configurations reveals clear performance trends. KnowProKD achieves favorable results over most compared methods from 1-shot to 16-shot settings, with more evident gains in higher shot scenarios (8–16 shots). This trend suggests that the proposed knowledge-aware prompts can benefit from

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TABLE II

CROSS-DATASET BENCHMARK EVALUATION. KNOWPROKD ACHIEVES CONSISTENTLY FAVORABLE PERFORMANCE ACROSS DIVERSE REMOTE SENSING DATASETS COMPARED WITH RECENT PROMPT-LEARNING BASELINES

	Source	Target							
	MLRSNet	PatternNet	RSSCN7	AID	RSICD	UCM	WHURS19	NWPU	*Average.*
CoOp	72.53	66.97	69.03	67.30	63.50	77.57	85.47	70.43	71.60
CoCoOp	71.70	65.67	68.80	66.63	62.57	76.40	85.33	70.30	70.92
ProGrad	70.83	62.97	69.93	67.80	61.10	75.13	81.23	69.10	69.76
MaPLe	76.83	68.53	71.43	65.13	59.53	79.90	85.23	72.80	72.42
PromptSRC	73.80	67.37	73.30	70.70	63.43	80.17	85.17	72.57	73.31
CoPrompt	77.83	70.50	75.33	73.00	66.33	80.87	89.30	73.87	75.88
DPLQ	78.73	68.17	70.60	66.70	62.27	79.93	91.07	73.13	73.83
PromptKD	75.67	78.63	72.93	73.50	64.97	71.10	85.77	73.63	74.53
KnowProKD	**79.83**	**78.93**	**76.87**	**74.39**	**68.83**	**80.93**	**93.27**	**76.70**	**78.72**

TABLE III

DOMAIN GENERALIZATION. PROMPT LEARNING METHODS ARE TRAINED ON IMAGENET AND EVALUATED ON DATASETS WITH DOMAIN SHIFTS

	Source	**Target**				
	China	Sydney	Helsinki	New York	Rio	Nairobi
CLIP	66.73	60.83	46.15	47.77	73.96	41.30
CoOp	71.51	64.20	47.99	49.71	75.21	55.1
Co-CoOp	71.02	64.07	48.75	50.63	76.18	56.2
MaPLe	70.72	64.07	49.15	**50.90**	76.98	62.30
KnowProKD	**72.79**	**65.49**	**49.43**	50.75	**77.25**	**65.7**

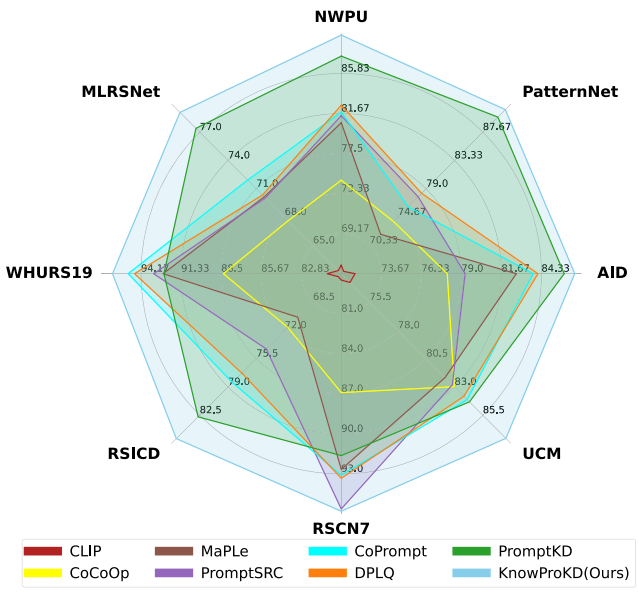


Fig. 4. Comparison with existing methods for base-to-novel generalization. KnowProKD obtains the highest HM on all eight remote sensing scene datasets.

the limited labeled samples available in few-shot learning while still preserving semantic guidance from external category knowledge. Compared with methods that mainly adapt class-name prompts, KnowProKD introduces attribute and description information to provide richer scene-level context, which is useful when only a few labeled examples are available for each category.

The observed gains can be attributed to three components: external knowledge generation, knowledge-aware prompt learning, and logits calibration. The generated attributes and descriptions provide complementary semantic cues, the knowledge-aware prompts inject these cues into the vision-language model, and the logits calibration module further refines predictions using attribute- and description-level evidence. Together, these components help the model distinguish visually similar but semantically different remote sensing scenes under limited supervision.

The performance advantages of KnowProKD are particularly evident in challenging remote sensing scenarios where domain-specific knowledge helps distinguish complex urban environments and natural landscapes. For example, scene categories with similar textures or spatial layouts may still differ in functional meaning, and external descriptions can provide useful semantic context for separating such categories. These findings support the practical value of incorporating external knowledge when labeled data are limited or costly to acquire. Furthermore, consistent improvements across different shot configurations indicate the adaptability of the proposed framework under varying data availability.

F. Visualization Analysis

1) Visualization of the CAM: In Fig. 6, we present a detailed visualization analysis using class activation maps (CAMs) generated from the penultimate layer of the image encoder, providing a comprehensive comparison between our proposed approach and baseline methods in the challenging base-to-novel transfer setting. The visualization employs red borders to delineate salient features within each image scene that demonstrate significant contribution to the final classification decisions. Our analysis focuses on four representative categories: “airplane” and “bridge” representing base categories that the model was initially trained on, while “ship” and “stadium” serve as novel categories to evaluate transfer learning capabilities. A thorough examination of the CAMs generated by the original CLIP model reveals notably weak perception of crucial discriminative features within the image domain. The baseline model’s attention patterns exhibit diffuse and unfocused activation, failing to precisely local-

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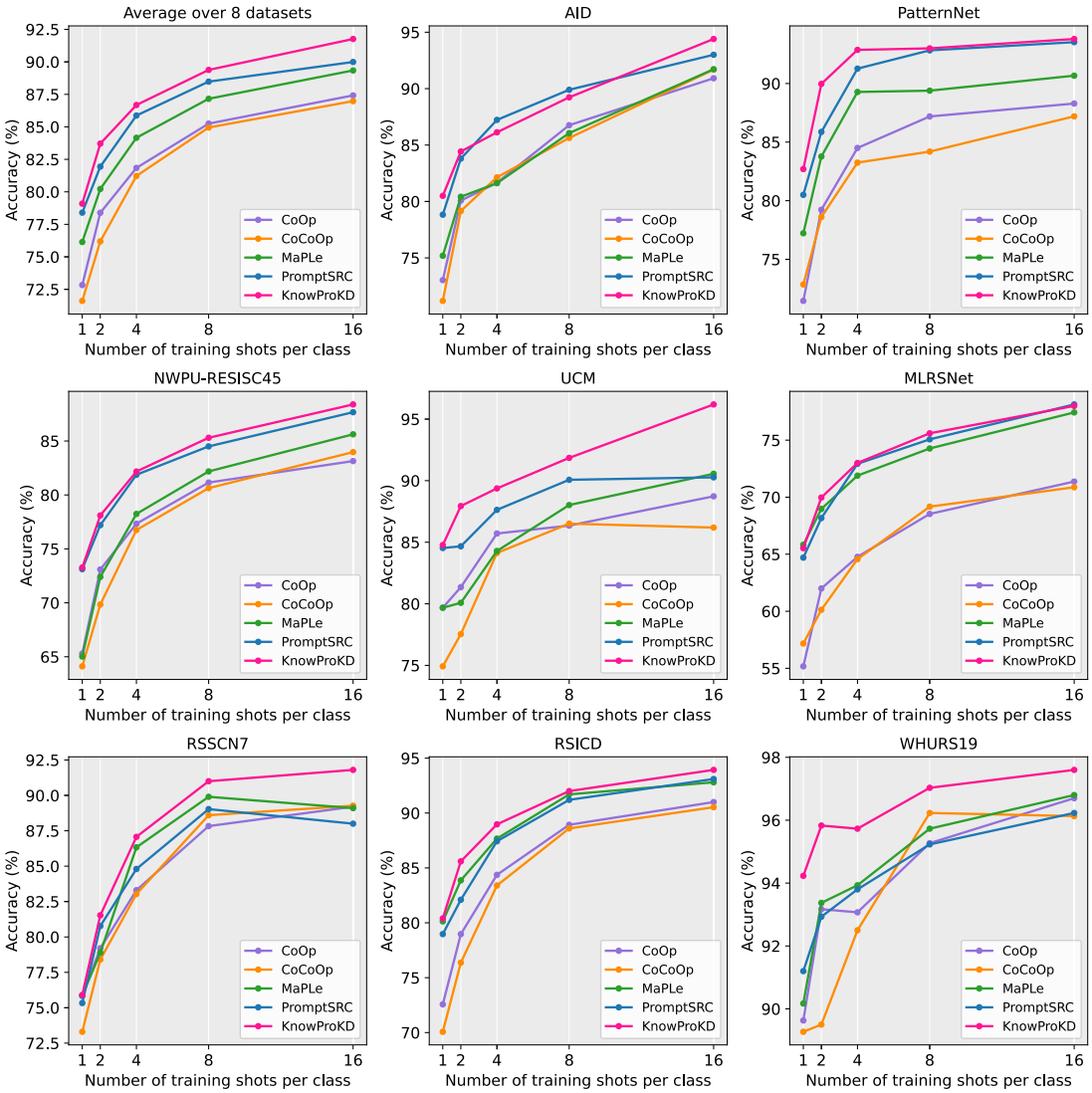


Fig. 5. Main results of few-shot learning experiments on eight remote sensing scene datasets. KnowProKD achieves favorable performance across different shot settings, showing the effectiveness of knowledge-aware prompt learning under limited training data.

ize category-specific attributes. Furthermore, when comparing against the standard fine-tuned approach, we observe that the attention directed toward category-relevant regions remains suboptimal. Even after conventional prompt learning, the activation patterns for target categories demonstrate insufficient specificity, with a tendency to disproportionately focus on background elements rather than discriminative foreground features. An exception appears in the stadium category, where improved attention localization is observed, though this may be attributed to the stadium's dominant spatial presence within the frame rather than enhanced feature learning. In contrast, the proposed knowledge-aware approach more clearly highlights semantically meaningful regions across diverse scenes. The model focuses more consistently on key foreground objects, indicating that the introduced knowledge cues help guide attention toward category-relevant features rather than background regions. For base categories such as airplane and bridge, the activated regions are more closely aligned with discriminative objects and structures. For novel categories such as ship and stadium, the model also maintains focused attention on visually and semantically relevant regions, suggesting improved base-to-novel transfer behavior. These observations indicate that the generated attributes and descriptions help the model form more transferable visual representations by encouraging attention to scene components that are useful for category recognition.

2) Visualization of the t-SNE Feature Map: As illustrated in Fig. 7, we compare the proposed knowledge-enhanced prompting method with the conventional category-only prompting baseline by projecting image features into a 2-D space using t-SNE. The visualization is conducted on three diverse remote sensing scene datasets to examine the learned feature distributions under different data characteristics. Compared with the baseline, KnowProKD produces more compact intraclass clusters and clearer interclass separation, indicating that diverse knowledge prompts help the model learn more discriminative representations. This effect is consistent with the quantitative results, where the proposed method improves both novel-class recognition and HM. Although interclass separation becomes more difficult when transferring from base to novel categories,

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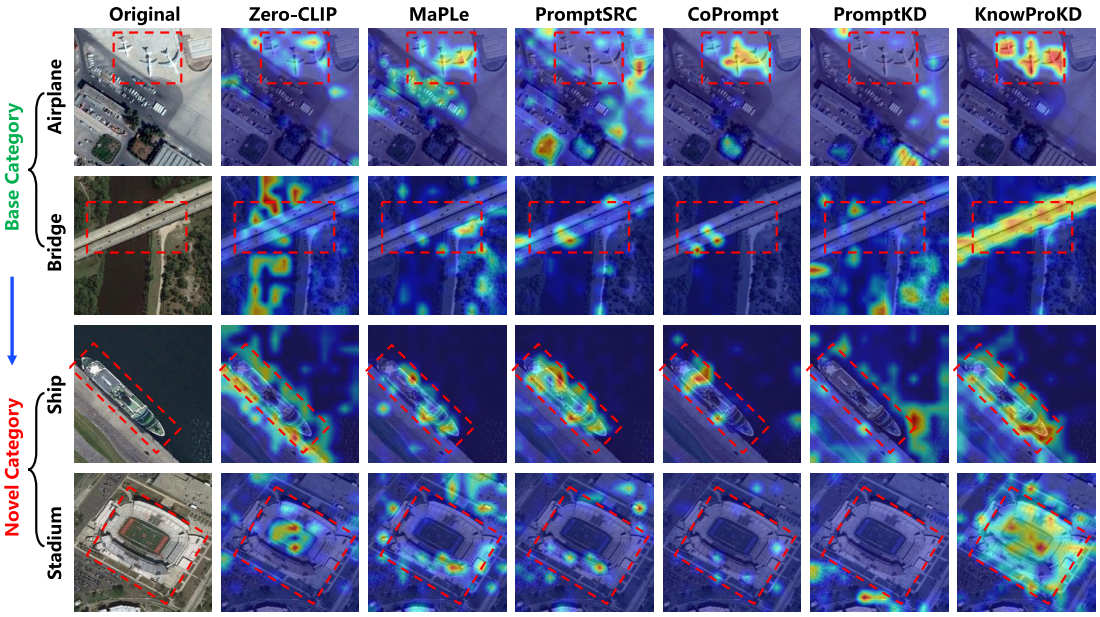


Fig. 6. Comparison of attention map visualization of the variant methods in the base-to-novel setting. Objects of focus are marked with red dashed boxes.

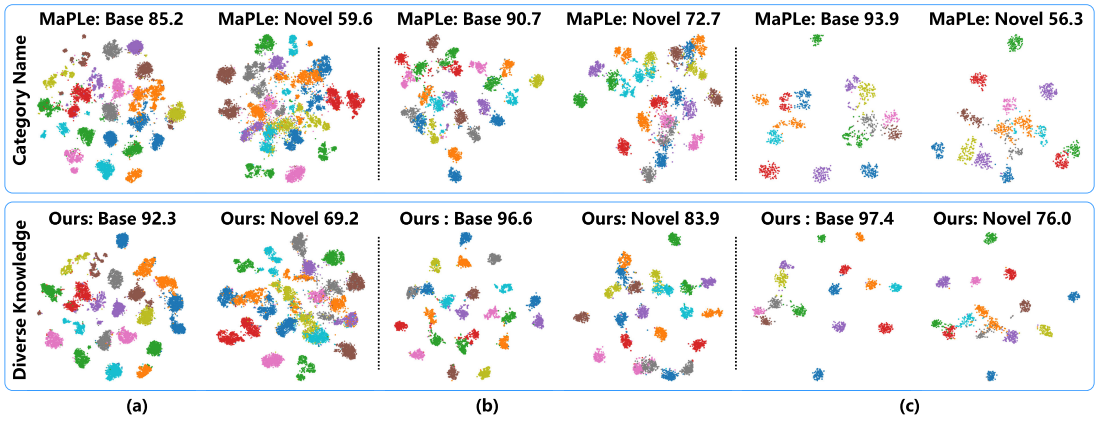


Fig. 7. t-SNE plots of image embeddings in the methods MaPLe and KnowProKD on three diverse remote sensing scene datasets. (a) MLRSNet. (b) NWPU-RESISC45. (c) RSICD.

KnowProKD still maintains visible classification boundaries for novel classes. These visualization results suggest that knowledge-enhanced prompts help the model learn more semantically meaningful representations that remain useful for both seen and unseen categories.

3) Representation Uniformity Analysis: To further substantiate that the proposed diverse knowledge logits calibration improves representation consistency across categories, we conduct a quantitative analysis of representation nonuniformity [48] on the MLRSNet dataset. Specifically, we measure *interclass distance* and *intraclass distance* of the learned features for both base and novel classes, where a larger interclass distance indicates better category separability and a smaller intraclass distance indicates tighter within-class compactness. As reported in Table IV, KnowProKD achieves higher interclass distances and lower intraclass distances compared with strong baselines, demonstrating a more balanced feature geometry under the base-to-novel setting. For example, compared with MaPLe, KnowProKD increases the interclass distance from 0.55 to 0.60 on base classes and from 0.48 to 0.55 on novel classes, while reducing the intraclass distance from 0.41 to 0.35 on base classes and from 0.43 to 0.39 on novel classes. These results provide quantitative evidence that the

TABLE IV

REPRESENTATION UNIFORMITY EVALUATION ON MLRSNET UNDER THE BASE-TO-NOVEL GENERALIZATION SETTING. B AND N INDICATE BASE AND NOVEL CLASSES, RESPECTIVELY. HIGHER INTERCLASS DISTANCE AND LOWER INTRACLASS DISTANCE ARE BETTER

Method	Inter-B ↑	Inter-N ↑	Intra-B ↓	Intra-N ↓
CoOp	0.55	0.47	0.42	0.44
CoCoOp	0.54	0.48	0.42	0.45
MaPLe	0.55	0.48	0.41	0.43
PromptSRC	0.56	0.49	0.40	0.43
CoPrompt	0.58	0.53	0.37	0.42
PromptKD	0.57	0.53	0.38	0.41
KnowProKD	**0.60**	**0.55**	**0.35**	**0.39**

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TABLE V

ABLATION OF PROPOSED COMPONENTS. CS, AP, AND KC REPRESENT THE KNOWLEDGE CONSTRAINTS, AWARE PROMPTS, AND KNOWLEDGE CALIBRATION, RESPECTIVELY

Attr Cs.	Desc Cs.	Attr AP.	Desc AP.	KC.	Base	Novel	HM
✗	✗	✗	✗	✗	93.12	71.71	80.42
✓	✗	✗	✗	✗	94.58	78.50	85.79
✗	✓	✗	✗	✗	94.91	77.02	85.03
✓	✓	✗	✗	✗	95.27	80.16	87.06
✓	✓	✓	✗	✗	95.53	81.05	87.70
✓	✓	✗	✓	✗	95.68	81.01	87.74
✓	✓	✓	✓	✗	95.84	81.62	88.16
✓	✓	✓	✓	✓	**96.39**	**82.91**	**89.14**

proposed method mitigates representation bias and improves generalization to novel categories.

G. Ablative Analysis

1) Effectiveness of Proposed Components: Our proposed KnowProKD framework consists of three key components: diverse knowledge-aware prompts, knowledge constraints, and diverse knowledge logits calibration. To quantify the contribution of each component, we conduct a comprehensive ablation study under the same base-to-novel generalization setting as the main experiments, and report the averaged results over eight remote sensing scene datasets in Table V.

We start from the MaPLe baseline, which achieves 93.12% base accuracy, 71.71% novel accuracy, and 80.42% HM. By introducing attribute-level or description-level knowledge constraints individually, the HM improves to 85.79% and 85.03%, respectively, indicating that both types of knowledge provide effective regularization for prompt tuning. When combining attribute and description constraints, the performance further increases to 87.06% HM, demonstrating their complementary benefits. Next, we incorporate diverse knowledge-aware prompts based on the constraint-enhanced setting. Adding only attribute-aware prompts or description-aware prompts leads to comparable improvements, yielding 87.70% and 87.74% HM, respectively. When both attribute-aware and description-aware prompts are enabled, the HM increases to 88.16%, showing that multigranularity knowledge prompts provide richer semantic cues for recognition. Finally, we activate the knowledge logits calibration module on top of all knowledge-aware prompts and constraints, yielding 96.39% base accuracy, 82.91% novel accuracy, and 89.14% HM. Compared to the MaPLe baseline, this corresponds to improvements of 2.56% on base accuracy, 11.20% on novel accuracy, and 8.72% on HM, validating the effectiveness and complementarity of the proposed components.

2) Knowledge Prompt Type: To determine whether performance gains stem from the semantic content of knowledge-aware prompts rather than the number of prompt branches, we conducted controlled experiments separating content and quantity effects. We first added extra prompt branches using only class names to assess the impact of quantity alone. Then,

TABLE VI

ABLATION ON KNOWLEDGE TYPES IN PROMPT LEARNING UNDER THE BASE-TO-NOVEL SETTING ON THE AID DATASET. PROMPTS #ID DENOTES DIFFERENT PROMPT BLOCKS, WHERE EACH BLOCK CORRESPONDS TO A SPECIFIC PROMPT TYPE CONFIGURATION

Prompts #1	Prompts #2	Prompts #3	Base	Novel	HM
class	-	-	95.00	71.07	81.26
class	class	class	95.36	74.82	83.85
class	attributes	class	95.81	76.29	84.94
class	class	descriptions	95.79	77.32	85.57
class	attributes	descriptions	**96.30**	**78.60**	**86.55**

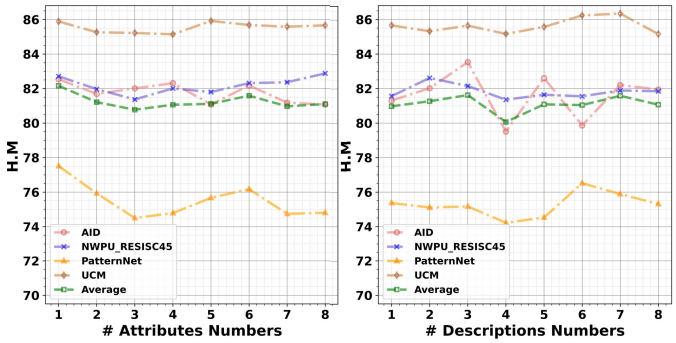


Fig. 8. Ablation study on the number of attributes (left) and descriptions (right) of each class prompt for prompt learning on the remote sensing scene datasets.

we introduced attribute or descriptive knowledge into the added branches to evaluate the effect of semantic enrichment. As shown in Table VI, increasing prompt quantity alone led to minor improvements. In contrast, prompts enriched with attributes or descriptions significantly boosted performance, especially on novel classes. Attribute-based prompts were more effective than descriptive ones, suggesting that structured attribute knowledge better supports visual-semantic alignment.

3) Knowledge Prompt Numbers: In the context of multiple knowledge-aware prompt learning, the quantity of injected knowledge emerges as a critical factor influencing model performance. To systematically investigate this relationship, we conducted extensive experiments examining how varying amounts of both attribute and descriptive knowledge affect model behavior and effectiveness. Our analysis reveals nuanced and nonlinear relationships between knowledge quantity and model performance. As illustrated in Fig. 8(a), when increasing the volume of attribute knowledge, we observed an initial slight decline in the fine-tuned model's performance, followed by a subsequent recovery and improvement phase. This pattern suggests a period of adaptation as the model learns to effectively integrate and leverage larger quantities of attribute information. Similarly, Fig. 8(b) demonstrates that while incorporating additional descriptive knowledge initially yields modest performance improvements, there exists a critical threshold beyond which further increases in knowledge quantity begin to negatively impact model performance. This performance degradation at higher knowledge volumes is particularly interesting and appears to be intrinsically linked to the qualitative aspects of the knowledge being incorporated. These

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TABLE VII

ABLATION ON PROMPT DEPTH UNDER THE BASE-TO-NOVEL SETTING ON THE AID DATASET. THE RESULTS SHOW THAT AN APPROPRIATE PROMPT DEPTH LEADS TO CONSISTENTLY BETTER PERFORMANCE FOR OUR METHOD

Depth	Base	Novel	HM
3	96.06	78.19	86.21
6	96.27	78.49	86.48
9	**96.30**	**78.60**	**86.55**
12	96.21	78.25	86.31

TABLE VIII

ABLATION ON LOGITS CALIBRATION COEFFICIENTS UNDER THE BASE-TO-NOVEL SETTING, AVERAGED OVER EIGHT REMOTE SENSING SCENE DATASETS. THE RESULTS INDICATE THAT APPROPRIATE COEFFICIENT VALUES CONSISTENTLY IMPROVE THE PERFORMANCE OF OUR METHOD

λ_1	λ_2	Base	Novel	HM
0.1	0.1	96.33	82.89	89.11
0.1	0.2	96.26	82.83	89.04
0.1	0.3	96.20	82.65	88.92
0.2	0.1	**96.52**	82.58	89.01
0.2	0.2	96.39	**82.91**	**89.14**
0.2	0.3	96.42	82.75	89.06
0.3	0.1	96.31	82.63	88.95
0.3	0.2	96.17	82.85	89.01
0.3	0.3	96.24	82.81	89.02

findings align remarkably well with established research in natural language knowledge injection [49], reinforcing the fundamental principle that the mere accumulation of knowledge is not sufficient for optimal performance. Instead, our findings highlight that the qualitative attributes of the incorporated knowledge, such as its relevance, accuracy, and coherence, are significantly more influential in determining the overall effectiveness of knowledge-aware prompt learning.

4) Knowledge Injection Depth: Following MaPLe [31], we investigate how the depth of knowledge-aware prompt injection affects the base-to-novel generalization performance. Since KnowProKD introduces multiple types of knowledge prompts, including attribute-based and description-based prompts, we evaluate different prompt depths to determine an appropriate injection configuration. As reported in Table VII, the overall performance varies moderately across different depths, and injecting prompts at depth 9 achieves the best results, with 96.30% base accuracy, 78.60% novel accuracy, and 86.55% HM. Using shallower depths, such as 3 and 6, leads to slightly lower HM scores, while injecting prompts too deep (e.g., depth 12) does not further improve performance. These results indicate that an intermediate depth provides a favorable balance for integrating diverse knowledge cues into the vision-language representation, and we adopt depth 9 as the default setting in all experiments.

5) Logits Calibration Coefficient: The diverse knowledge logits calibration module contributes to the performance gain by fusing the logits predicted from class prompts, attribute prompts, and description prompts. In our design, the class-prompt logits serve as the primary prediction signal, while the attribute- and description-prompt logits provide complementary knowledge evidence for calibration. Accordingly, the final prediction is obtained by a weighted combination controlled by λ_1 and λ_2, where λ_1 and λ_2 denote the weighting coefficients of the attribute and description logits, respectively. As summarized in Table VIII, we evaluate different settings of λ_1 and λ_2. We observe that the best HM is achieved when $\lambda_1 = \lambda_2 = 0.2$. Meanwhile, the overall performance remains relatively stable under moderate changes of λ_1 and λ_2, indicating that the proposed calibration strategy is not sensitive to the exact choice of these weights. Therefore, we use fixed coefficients in all experiments for simplicity and reproducibility, and the performance gain mainly comes from the diverse knowledge evidence introduced by attribute and description prompts rather than careful weight tuning.

TABLE IX

COMPARISON OF EFFICIENCY AND PERFORMANCE UNDER DIFFERENT PROMPT LEARNING METHODS. FOLLOWING MAPLE [31], FPS IS MEASURED ON A SINGLE NVIDIA A100 80 GB GPU WITH BATCH SIZE 100 UNDER THE SAME INFERENCE PROTOCOL

Method	Params (M) ↓	FPS ↑	HM ↑
CoOp	**0.002**	1344	78.03
CoCoOp	0.354	15.08	78.43
MaPLe	0.41	1365	80.42
PromptSRC	0.046	1380	81.62
PromptKD	0.046	1710	86.67
KnowProKD	0.41	**1760**	**89.14**

6) Computational Efficiency: As shown in Table IX, we compare different prompt learning methods in terms of efficiency and performance, including trainable parameters, inference speed (FPS), and HM. Following MaPLe [31], FPS is measured on a single NVIDIA A100 80 GB GPU with a batch size of 100, and is computed as the number of images processed per second during model inference under the same inference protocol. CoOp and CoCoOp exhibit relatively limited performance gains, while CoCoOp also suffers from a significantly lower inference throughput due to its additional conditioning module. In contrast, PromptSRC achieves a better tradeoff with only 0.046-M trainable parameters and competitive efficiency. More importantly, PromptKD further improves HM to 86.67% while maintaining high FPS, demonstrating the effectiveness of prompt distillation. Building upon this, KnowProKD attains the highest HM of 89.14% in this comparison while using 0.41 M trainable parameters and maintaining fast inference speed. These results indicate that KnowProKD provides a favorable accuracy-efficiency tradeoff compared with prior prompt tuning baselines.

H. Cross-Region Urban Land-Use Mapping and Statistics Analysis

1) Cross-Region Urban Land-Use Mapping: The base-to-novel and cross-dataset experiments in Sections IV-B and IV-C demonstrate the scene transfer capability of the proposed KnowProKD framework. To further examine its practical utility, we conducted large-scale cross-regional urban

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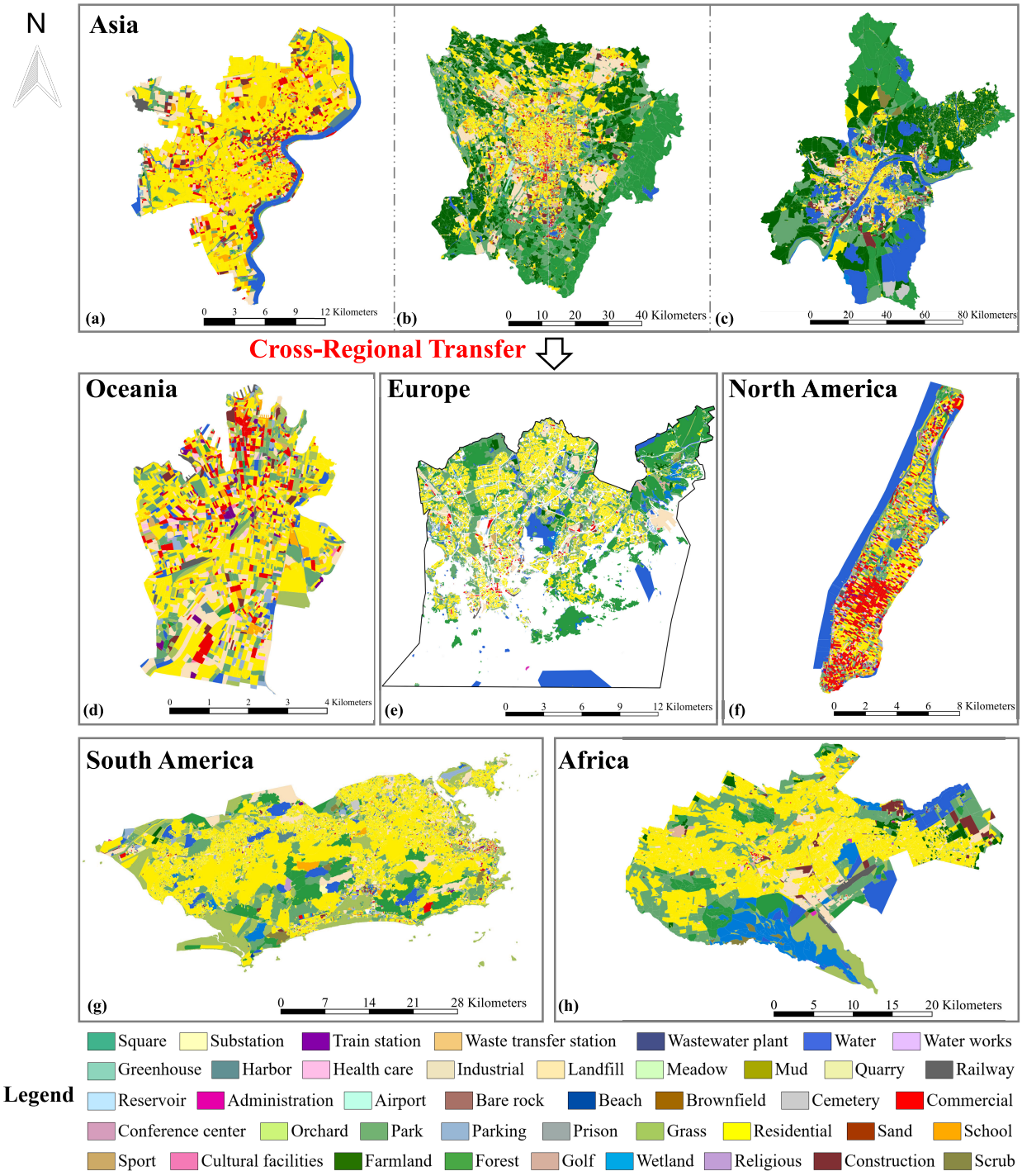


Fig. 9. Cross-regional urban land-use mapping results on the GUN dataset. The model is trained on China region and then transferred to other global cities. (a) Shanghai. (b) Chengdu. (c) Wuhan. (d) Sydney. (e) Helsinki. (f) New York. (g) Rio. (h) Nairobi.

land-use mapping experiments using the GlobalUrbanNet (GUN) dataset [50]. The model is trained with diverse knowledge prompting using data from China region of GUN and then transferred to representative metropolitan areas across five other continents. This cross-regional deployment setting is challenging because urban morphology, land-use composition, building density, and functional organization can vary substantially across continents. It therefore provides a practical test of whether the learned knowledge-aware prompts can support transfer to urban environments that differ from the source training region.

The CLIP-based architecture of KnowProKD allows flexible adaptation of classification taxonomies, which is useful for handling heterogeneous urban functions and land-use patterns across different global regions. Urban areas across continents exhibit variations in development density, architectural styles, cultural influences, and spatial organization. For example, high-density Chinese cities often contain compact mixtures

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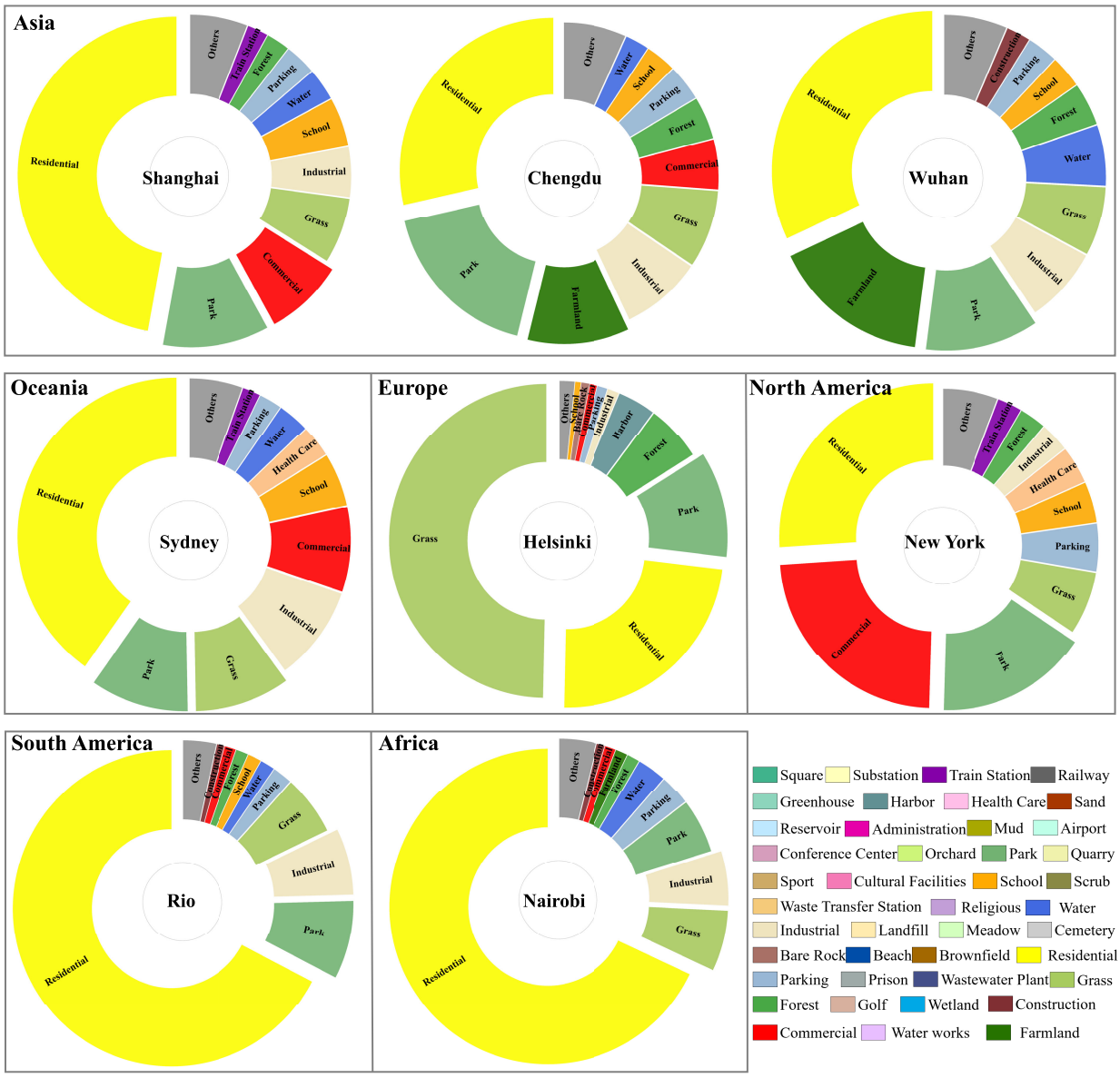


Fig. 10. Donut chart of urban land use statistics in different continents. The model is trained in China and then transferred to other global cities for open-vocabulary scene classification.

of commercial, residential, and industrial areas, while cities such as Helsinki and Nairobi include more prominent natural features such as forests, water bodies, and open spaces. These differences make cross-regional land-use mapping more difficult than within-region evaluation, because the model must transfer scene knowledge across both visual appearance and functional land-use patterns.

As demonstrated in Fig. 9, KnowProKD is first deployed for urban land-use classification in Shanghai, Chengdu, and Wuhan, and is then applied to diverse global cities. The mapping results indicate that the model can transfer from high-density Chinese urban patterns to different urban contexts in Europe, North America, Africa, South America, and Oceania. This cross-context evaluation covers both highly developed cities and emerging urban regions, providing evidence that KnowProKD can produce meaningful land-use maps under different urban typologies and development stages. These results further illustrate the potential of knowledge-aware prompt learning for practical urban monitoring and cross-regional remote sensing applications.

2) Global Urban Land-Use Statistics Analysis: As illustrated in Fig. 10, the donut charts summarize the land-use distributions inferred by KnowProKD after transferring the model trained on China region of the GUN dataset to representative global cities. The results show that residential areas occupy a large proportion in most cities, especially in Shanghai, Rio de Janeiro, and Nairobi. Green spaces, including parks and grasslands, are more prominent in Helsinki, Sydney, and Chengdu, reflecting different urban ecological conditions and planning priorities. In contrast, high-density cities such as Shanghai and New York show more compact urban compositions, with relatively stronger commercial or built-up components. Industrial areas are more evident in cities such as

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Sydney, Chengdu, and Wuhan, while developed cities such as Helsinki and New York contain more diverse categories related to transportation, public services, and institutional facilities.

These observations indicate clear regional differences in urban land-use structure, which are associated with variations in urban density, economic development, planning strategies, and geographic conditions. Compared with cities that show more mixed land-use patterns, Rio de Janeiro and Nairobi present relatively concentrated distributions dominated by residential and green-space categories. Overall, the statistical results complement the mapping results in Fig. 9, showing that KnowProKD can provide interpretable cross-regional land-use estimates and has potential value for global urban land-use monitoring.

V. Conclusion

In this article, we have presented KnowProKD, a knowledge-enhanced prompt learning framework that addresses limitations in existing vision-language model adaptation approaches. While prompt learning has emerged as a useful paradigm for fine-tuning foundation models like CLIP for downstream tasks, conventional methods that rely solely on image category information often struggle with limited generalization capabilities and task-specific overfitting. Our work makes several contributions to knowledge-guided prompt learning for remote sensing scene classification. First, we introduce a sophisticated knowledge integration mechanism that leverages LLMs to generate rich semantic descriptions and attribute relationships, moving beyond simple category labels. This external knowledge injection provides the model with a more comprehensive understanding of visual concepts and their relationships. Second, we develop a novel constrained prompt learning framework that explicitly enforces knowledge-based constraints during training, helping to regularize the feature space and prevent overfitting. Third, we propose a diverse knowledge logits calibration module that operates exclusively during inference, effectively balancing the model's predictions across both base and novel categories.

Through empirical evaluation across multiple remote sensing scene recognition benchmarks, we demonstrate that KnowProKD achieves favorable performance compared with existing prompt learning approaches. The framework obtains clear gains in cross-dataset and few-shot learning scenarios, supporting its generalization capability. The visualization analysis further confirms that the knowledge-enhanced approach leads to more semantically meaningful and discriminative feature representations. Looking forward, we believe this work opens up promising directions for future research in knowledge-augmented visual understanding. Potential extensions include exploring more sophisticated knowledge integration mechanisms, investigating dynamic knowledge selection strategies, and applying our framework to other vision-language tasks. The principles developed in this work could also inform the design of more robust and generalizable foundation models for computer vision applications.

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